

Long-Term Evolution of Lunar PNT Constellation Performance

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Project Motivation:

Lunar exploration is going from simple exploration to sustained lunar evolution:

- In-situ resource exploitation/manufacturing
- Advanced infrastructure
- Sustained habitation and exploration
- Increase in autonomy and independence of complex engineering vehicles

This lunar exploration/sustainability requires precise navigation systems:

- Solution: Positioning systems (GPS) that maximize performance over an extended period of time, even without station-keeping or other additional support

How do we use natural phenomena to help us achieve these requirements?

- Exploit N-body dynamics to find stable/exploitable positions
- Analyze how perturbations affect the coverage of critical areas

Literature and Research Background:

The following sources were used as background information when conducting the study:

- Orbits were selected based on on a mixture of recommendations taken from Keidai Ilyama's paper *Trade-off Analysis for Lunar Augmented Navigation Service (LANS) Constellation Design* and additional stability testing of various inclination/SMA combos via MATLAB
- Some orbital dynamics data was taken from JPL's Solar System Database, and SPICE was used for some calculations
- DOP calculations/formulas were taken from the *GPS Overview AA278: Lunar Positioning, Navigation, and Timing Lecture #2* with code implementation and planning taken from *AA278_2026_HW1_P1*.
- DOP context/insight was gathered from papers such as *Dilution of Precision-Based Lunar Navigation Assessment for Dynamic Position Fixing* from NASA Goddard.
- Additional information, such as PNT context and terminology, was taken from various lectures, talks, and discussions during the course.

Locations of Interest:

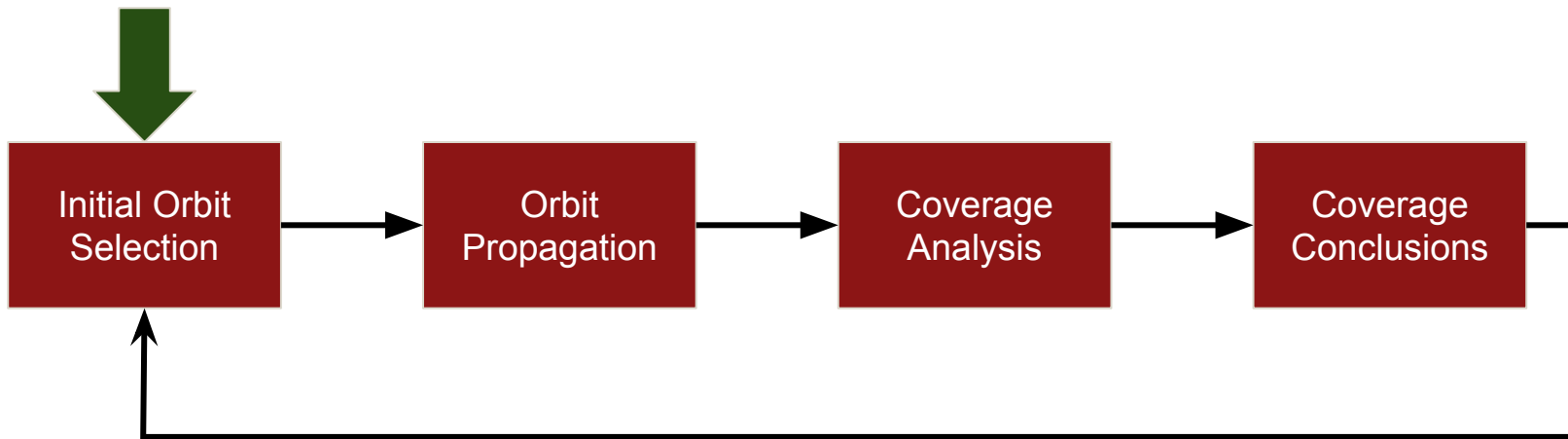
There are four main locations that NASA has identified as favorable in the South Pole:

- Nobile Rim 2:
 - Prime candidate spot for Artemis IV
 - Flat terrain, ice
 - -84.2° Latitude, 60.03° Longitude
- Malapert Massif:
 - Lunar mountain with elevation $\sim 5,000$ meters relative to base
 - Near-perpetual sunlight
 - -86.0° Latitude, 2.06° Longitude
- Henson SSE Rima Crater:
 - Roughly 43 km wide and 4 - 5 km in depth (Crater)
 - Near permanently shadowed regions and high elevation ridges (good sunlight)
 - -89.1° Latitude, -109.3° Longitude
- Faustini Rim A:
 - Roughly 41 km wide and near Faustini Crater
 - Ancient valleys with great potential for scientific discovery (mineral variety)
 - -87.3° Latitude, 77.0° Longitude

Analysis Pipeline - Checkpoint 1:

Initial Orbit Selection:

- All analysis starts with the first “guess” of what a good orbit is



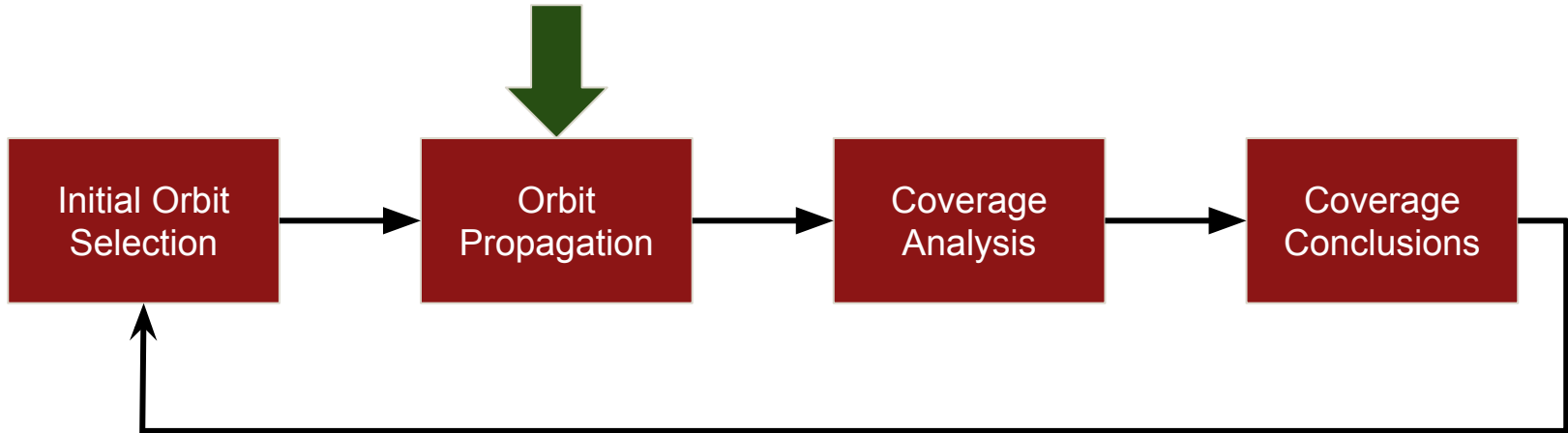
Initial Orbit Selection

- The orbits used for this study are all Elliptical Lunar Frozen Orbits (ELFOs), with arguments of periapsis set to 90°, maximizing dwell time over the south pole
 - Semi-major axes were selected as whole number ratios of the lunar sidereal period
 - 10:1, 20:1, and 40:1 (10%, 5%, and 2.5%) respectively
 - The lunar frozen orbit condition is defined based on eccentricity and inclination - for a given inclination, the necessary eccentricity is as follows:
 - $e = (1 - (5/3) \cos^2(i))^{1/2}$
 - Inclinations below 40 degrees are unstable as the eccentricity becomes non-real
 - At inclinations above 60 degrees, the necessary eccentricity becomes so large that orbits intersect the lunar surface for all reasonable semi-major axes
 - Inclinations are calculated in the Earth-Moon orbit plane (OP) frame and converted to Moon-Centered Inertial (MCI) for propagation
- The constellations are all 12/3/1 Walker constellations
 - 3 planes of 4 equally-spaced satellites each, for a total of twelve satellites
- Epoch time is 12:00 am UTC on 06/01/2026

Analysis Pipeline - Checkpoint 2:

Orbit Propagation:

- Orbit is propagated in time to analyze effects of perturbations and disturbances



Orbit Propagation:

- Orbit propagation was performed numerically in MATLAB, using *ode113*
- Initial orbits were defined for each satellite using orbit elements, then converted to MCI position/velocity vectors for propagation
- Perturbations covered in propagation:
 - Lunar Gravity
 - Lunar J_2 Effects
 - Earth Third-Body Perturbation
 - Solar Third-Body Perturbation
- Orbits were propagated for 1 year, using 60 second timesteps, with the solver terminating early for a satellite if it either impacted the lunar surface or left the moon's Hill sphere

```
%acceleration due to Moon's oblateness
coef = - 0.5 * GM_Moon * J2_Moon * R_Moon .^ 2;
term = 3.0 / r_Sat_Moon .^ 4 - 15 * (z_Sat .^ 2 / r_Sat_Moon .^ 6);

ax_J2 = coef .* term .* x_Sat ./ r_Sat_Moon;
ay_J2 = coef .* term .* y_Sat ./ r_Sat_Moon;
az_J2 = coef .* (6 * z_Sat / r_Sat_Moon .^ 5 + term * z_Sat ./ r_Sat_Moon);

a_J2_pa = [ax_J2; ay_J2; az_J2];

%Rotate from PA to MCI
a_J2 = rot_mci_to_pa'*a_J2_pa;

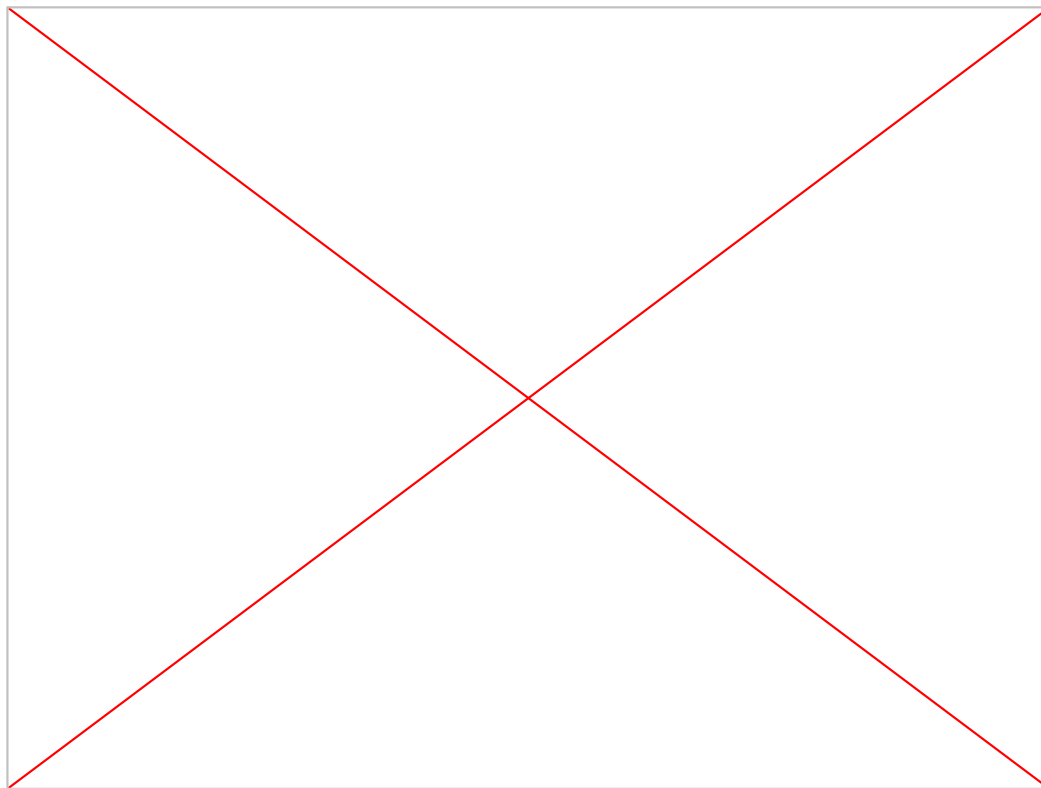
%Impact due to Earth's 3-body gravity
pos_Earth_Moon = cspice_spkpos('Earth',t,'J2000','NONE','MOON');
pos_Sat_Earth = pos_Earth_Moon - r_vec(1:3);

a_Earth = GM_Earth*(pos_Sat_Earth/norm(pos_Sat_Earth)^3 - ...
pos_Earth_Moon/norm(pos_Earth_Moon)^3);

%Impact due to Sun's 3-body gravity
pos_Sun_Moon = cspice_spkpos('Sun',t,'J2000','NONE','MOON');
pos_Sat_Sun = pos_Sun_Moon - r_vec(1:3);

a_Sun = GM_Sun*(pos_Sat_Sun/norm(pos_Sat_Sun)^3 - ...
pos_Sun_Moon/norm(pos_Sun_Moon)^3);
```

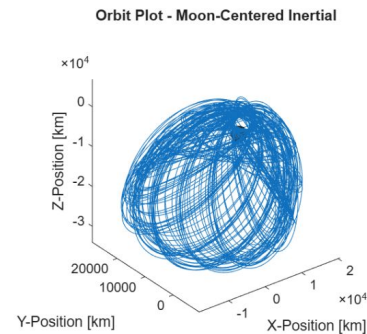
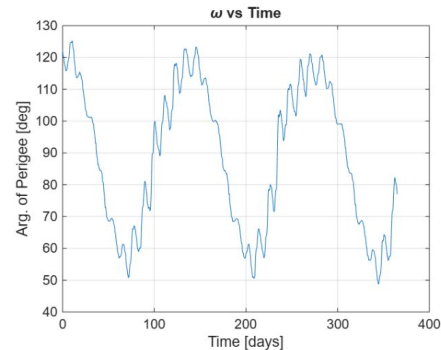
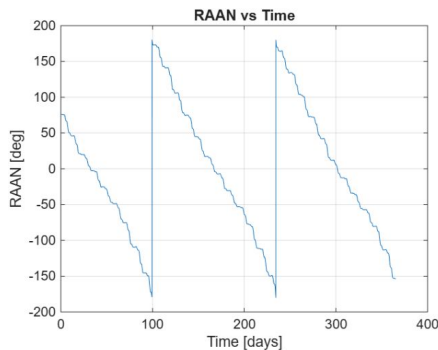
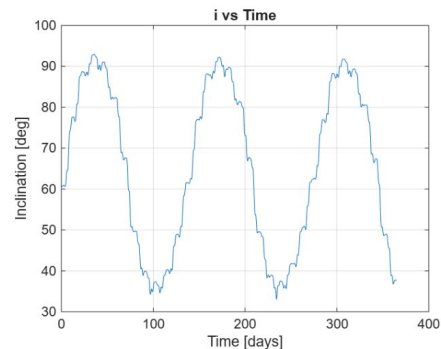
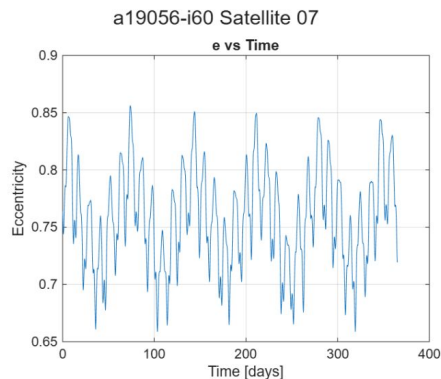
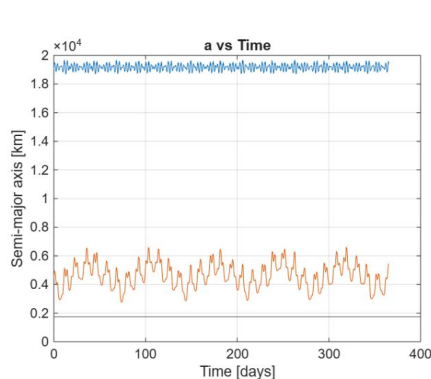

Orbit Propagation Animation



Orbit Propagation Conclusions

- In general, the majority of constellations remained intact throughout the year-long analytics window
 - One exception - the 60° constellation for the lowest semimajor axis impacts the lunar surface 3-4 days post-epoch, as variations in eccentricity lower the periapsis below the surface
- Nodal precession rotates the orbits clockwise as expected (due to the J_2 Coefficient)
- Lidov-Kozai mechanism causes large oscillations in inclination, ranging from 35-90°, although somewhat exaggerated by conversion between OP and MCI
- Full apsidal precession is prevented by the frozen orbit, and instead, the periapsis oscillates within 30-40° of 90°.
- While the orbits do not remain consistent, they remain relatively constant w.r.t each other, and maintain characteristics that ensure good visibility & dwell time over the south pole

Orbit Propagation Conclusions

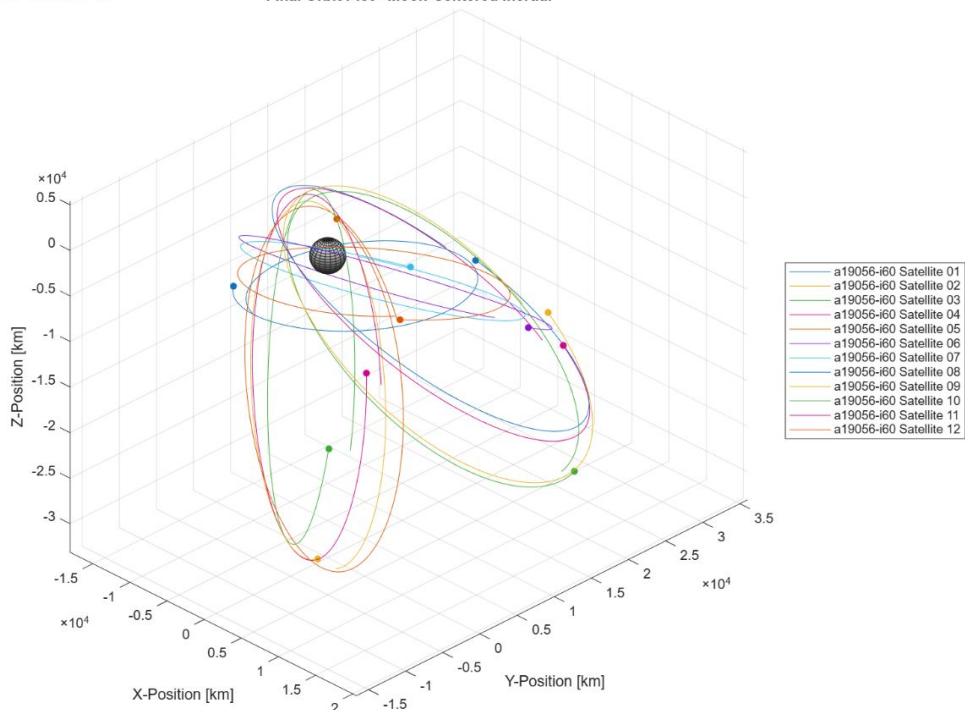
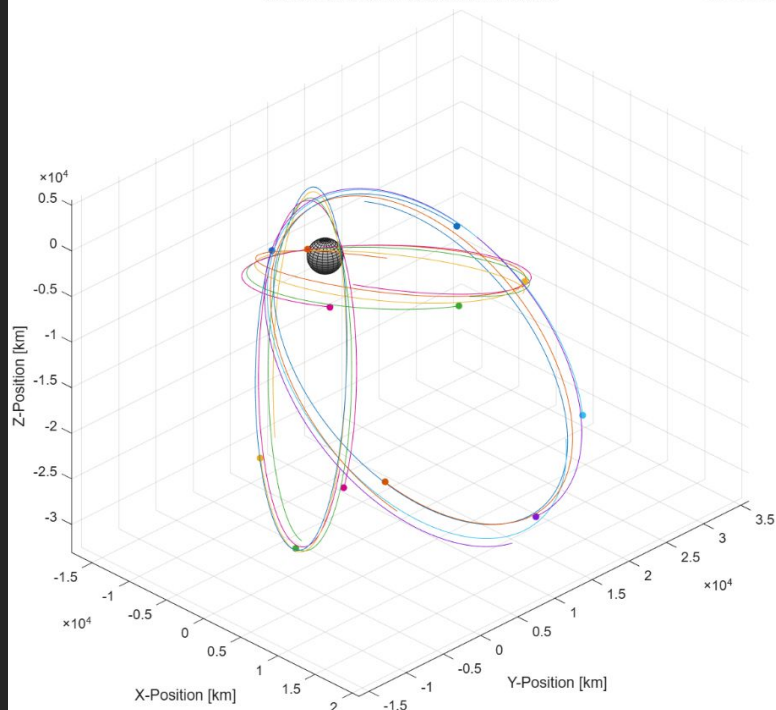


12 Orbit Propagation Conclusions

Initial Orbit Plot - Moon-Centered Inertial

a19056-i60 Satellite 12

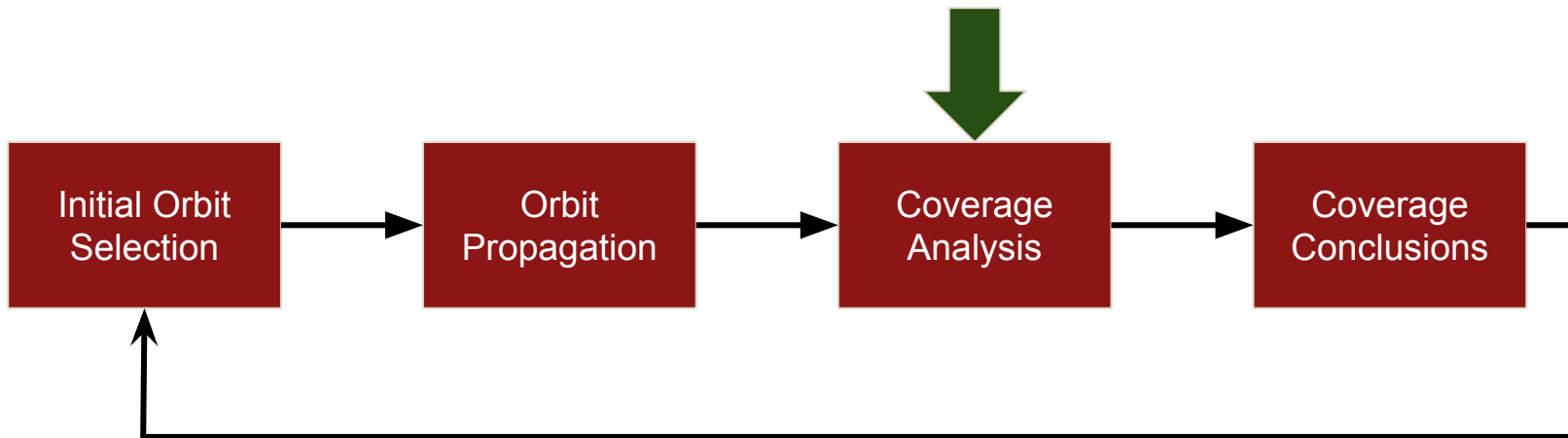
Final Orbit Plot - Moon-Centered Inertial



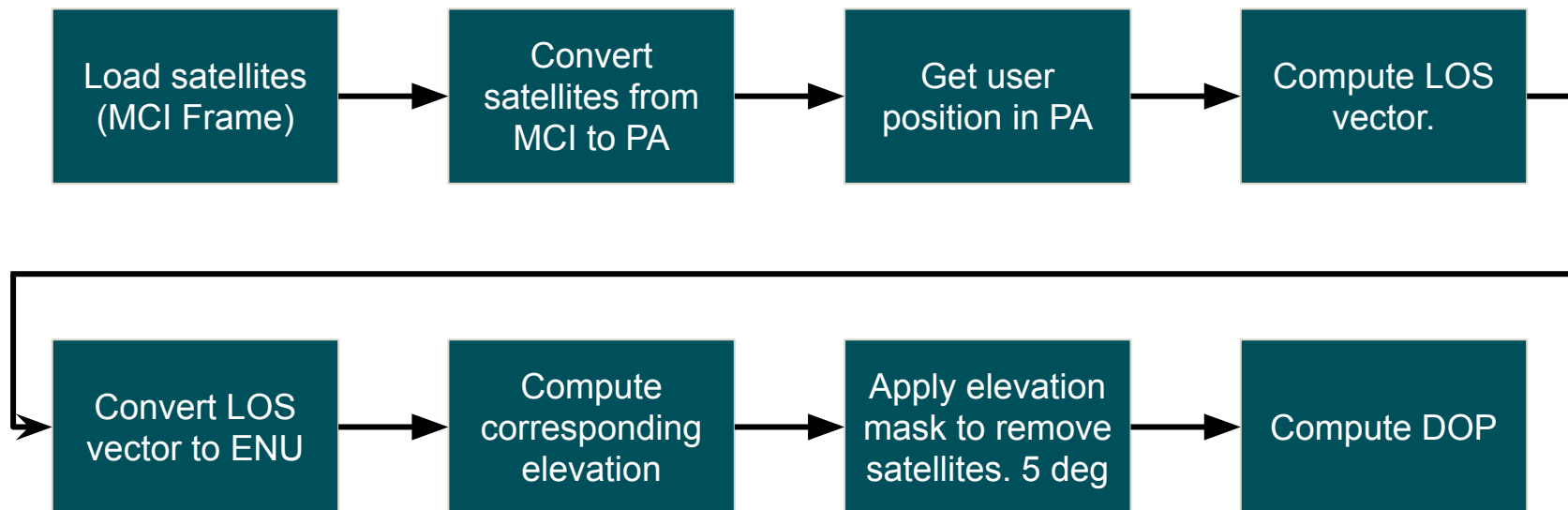
Analysis Pipeline - Checkpoint 3:

Coverage Analysis:

- Simulations are run to determine the coverage at target location given the selected constellation



Coverage Analysis - Pipeline:



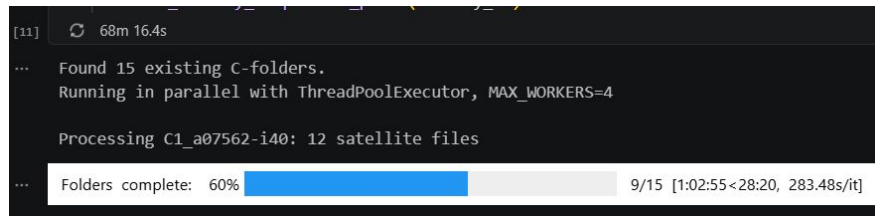
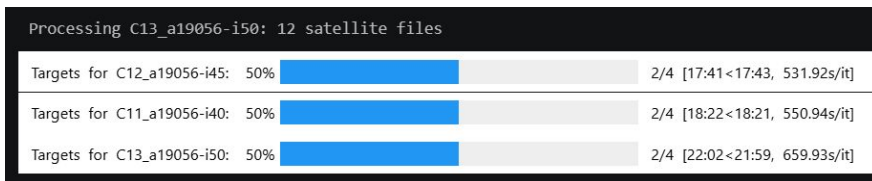
Coverage Analysis - Code:

Code structure:

- Jupyter notebook with functions from previous homeworks as motivation
- Modular - Allows for easy changes in plotting
- CSV input -> CSV output

Simulation size:

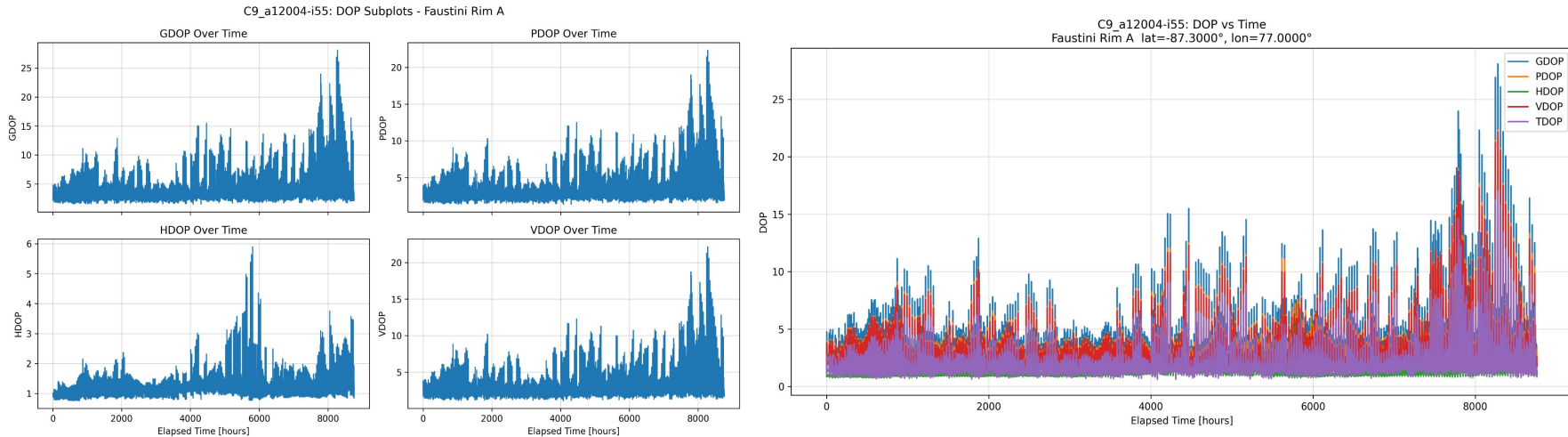
- 15 unique simulations. Each simulation includes 4 locations. Each location includes 4 plots. **240 total plots + Residual Data**
- Multi-hour run for each analysis



Coverage Analysis - Example Output:

File C9_a12004-i55 for Faustini Rim A Location

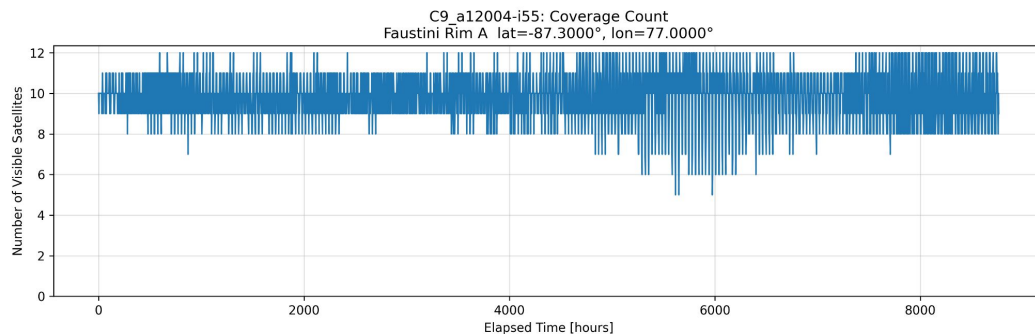
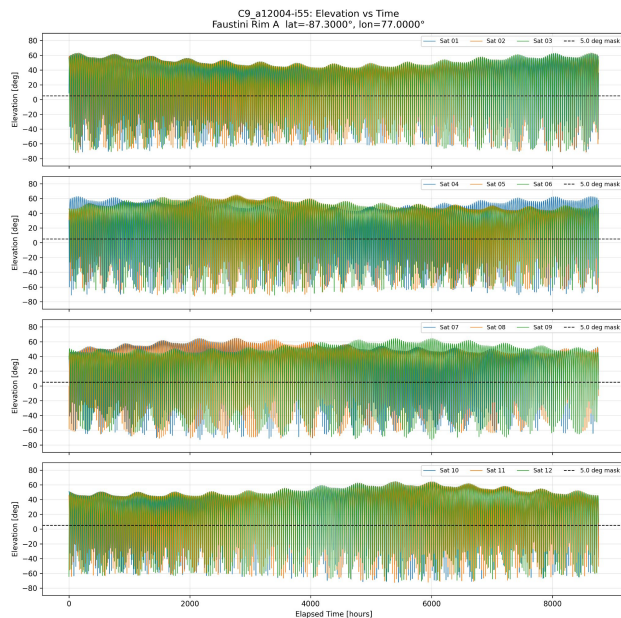
- DOP Over Time



Coverage Analysis - Example Output 2:

File C9_a12004-i55 for Faustini Rim A Location

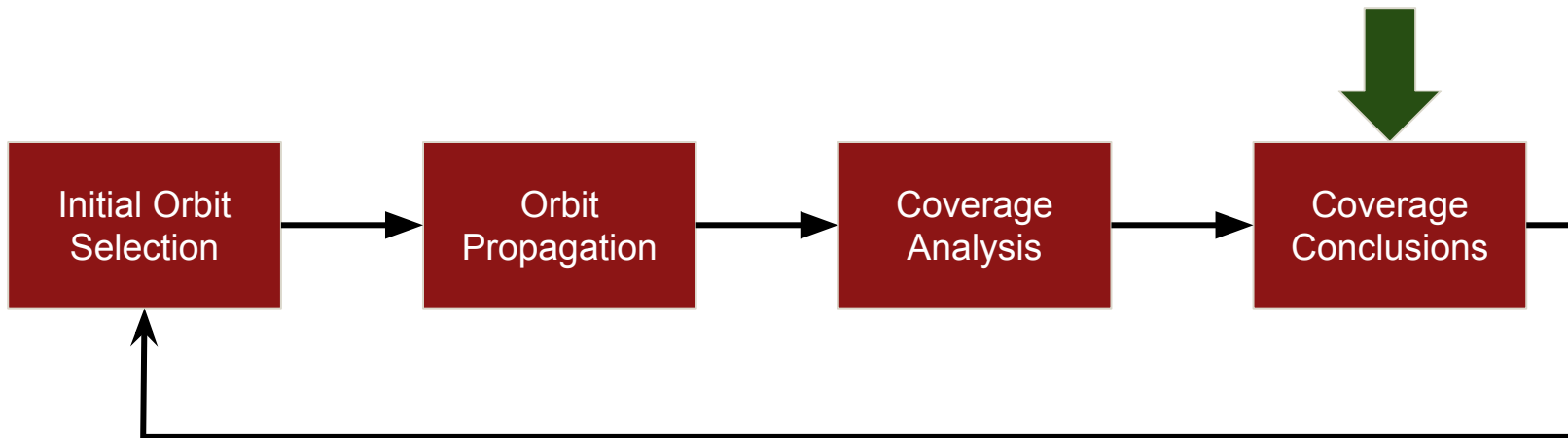
- Elevation and Satellite Count



Analysis Pipeline - Checkpoint 4:

Coverage Conclusions:

- Remarks regarding whether the coverage is adequate for selected location



Coverage Conclusions - Analysis Metric:

There was lots of data to parse, so there needs to be a rubric:

1. An optimal location must have 4 satellites in view:
 - x, y, z, and time
2. Must have good DOP values

Geometric Dilution of Precision
(GDOP)

Position Dilution of Precision
(PDOP)

Horizontal Dilution of Precision
(HDOP)

Vertical Dilution of Precision
(VDOP)

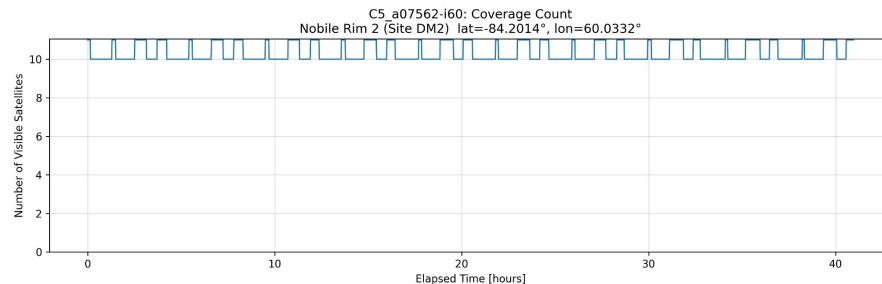
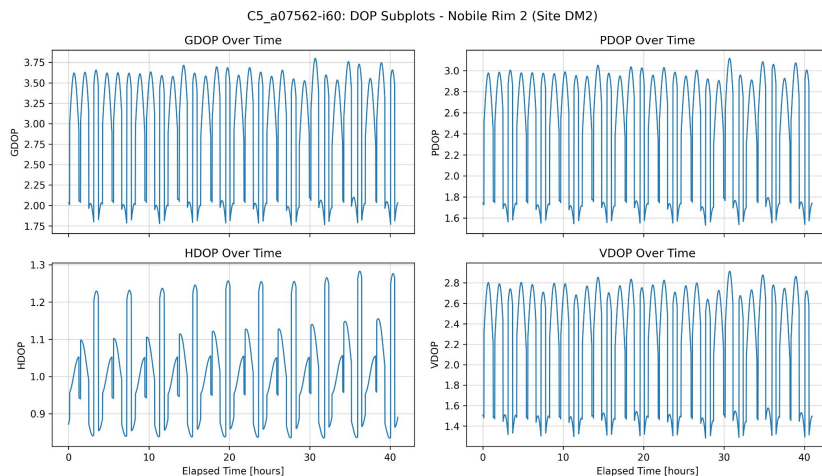
DOP Value	Rating ^[5]	Description
< 1	Ideal	Highest possible confidence level to be used for applications demanding the highest possible precision at all times.
1–2	Excellent	At this confidence level, positional measurements are considered accurate enough to meet all but the most sensitive applications.
2–5	Good	Represents a level that marks the minimum appropriate for making accurate decisions. Positional measurements could be used to make reliable in-route navigation suggestions to the user.
5–10	Moderate	Positional measurements could be used for calculations, but the fix quality could still be improved. A more open view of the sky is recommended.
10–20	Fair	Represents a low confidence level. Positional measurements should be discarded or used only to indicate a very rough estimate of the current location.
> 20	Poor	At this level, measurements should be discarded.

[Source](#)

Coverage Conclusions - Good Example:

Excellent location: **Nobile Rim 2** for constellation **a07562-i60**

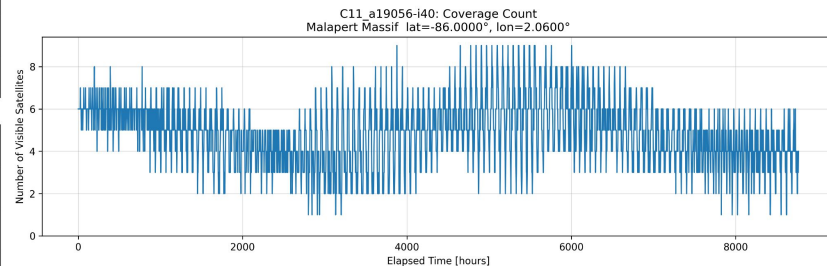
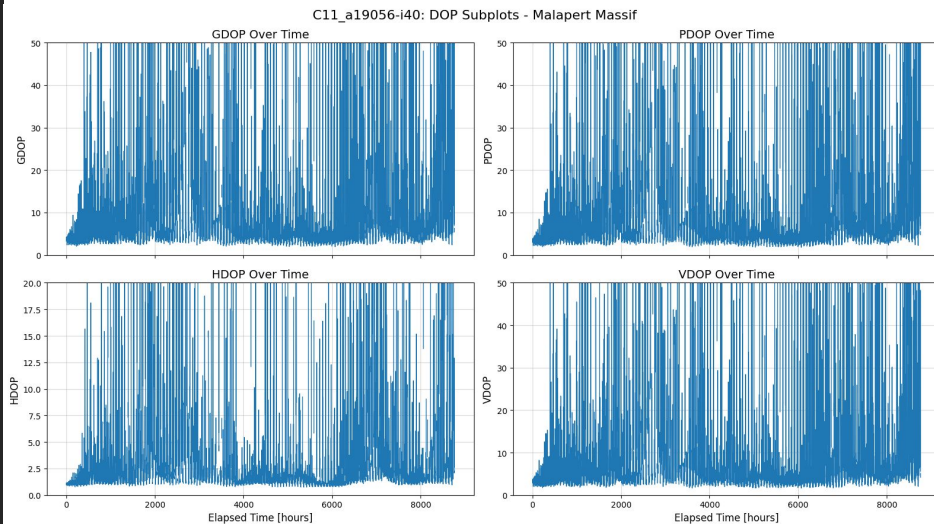
- Always 10+ satellites and DOP remains bounded and oscillatory
- Note: *crashes after only 40 hours*



Coverage Conclusions - Bad Example:

Bad location: **Malapert Massif** for constellation **C11_a19056-i40**

- Dips to only 1 satellite worth of coverage - especially prone to the orbital perturbations
- Extended periods of time where DOP is exceedingly large



Coverage Conclusions - Summary:

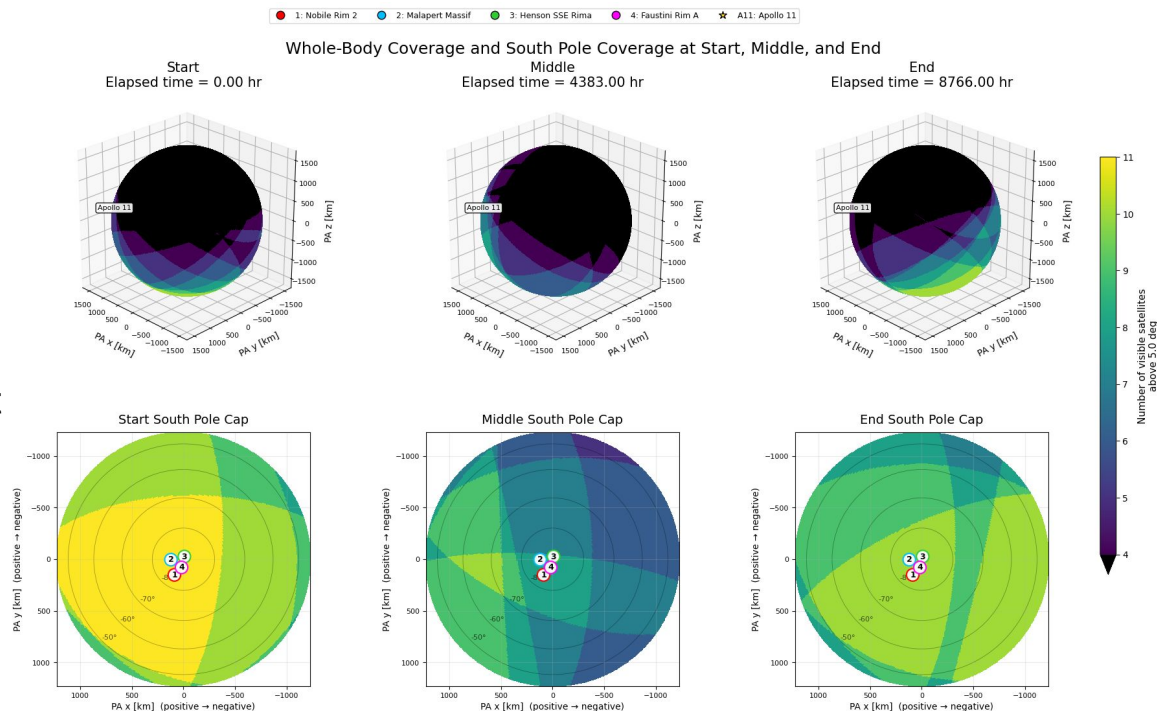
Location	Best Constellation	Satellite Coverage	DOP Average
Nobile Rim 2	a19056-i60	10.61	2.72
Malapert Massif	a19056-i60	10.61	3.31
Henson SSE Rima Crater	a19056-i60	10.62	2.70
Faustini Rim A	a19056-i60	10.62	2.71

Note: This clustering makes intuitive sense because locations are clustered near each other

Coverage Conclusions - 3D

What if lunar inhabitants need to explore:

- 0.5 degree step size
- Includes the current locations of interest + Apollo 11 site for reference
- Expensive to compute but gives good insight into coverage evolution
- Apollo 11 site explorable $\frac{1}{2}$ year into simulation



Constellation Conclusions:

In general, we were able to make the following conclusions:

1. Although a variety of inclination and semimajor axes combinations were analyzed, the best case was found to be **a19056-i60**, which worked best for all target locations, providing the best DOP and maintaining long-term stability. This is not unsurprising as the high inclination (60°) ensures that satellites will be higher in the sky over the polar regions, while the large semimajor axis (19,056 km) leads to long dwell times, and prevents any risk of surface collisions even as the orbit is perturbed
2. The constellation **a07562-i60**, although promising at the beginning, crashed within days of the simulation starting, as Lidov-Kozai pumped eccentricity and lowered the periapsis below the lunar surface, while **a12004-i60** simply had worse DOP performance than **a19056-i60**. Lower inclination constellations had increasingly worse DOP than any of the 60° constellations

Important Acknowledgments and Future Work:

- Assumptions/future work for orbit propagation:
 - Didn't include higher-order gravity terms for Moon, which likely have non-negligible impact. Well worth investigating with additional time
 - Didn't factor in impacts of solar radiation pressure on satellites
 - Didn't consider the possibility of station-keeping - satellites were assumed to be free-flying from epoch
- Assumptions/future work for coverage:
 - No elevation considerations. May improve/worsen certain locations - especially near craters or mountains
 - Did not estimate error in positioning. May be worth investigating with more time
 - With more compute, full evolution of coverage (animation), may give visual insight into coverage changes for all locations